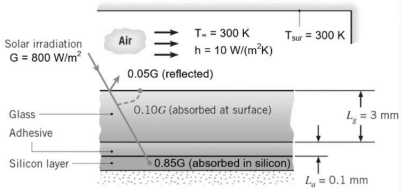
1. (40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass (k_g = 1.5 W/m·K), a 0.1-mm-thick optical grade adhesive (k_a = 1 W/m·K), and a very thin layer of silicon within which solar energy is converted to electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si}, as described by the expression $\eta = a - bT_{Si}$, where a = 0.553 and b = 0.001 K^{-1}. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4}$ m^2·K/W. There is no contact resistance between the glass and the adhesive.



In a sunny day, there is G = 800 W/m^2 solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part 6f the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of T_∞ = 300 K and convective heat transfer coefficient of h = 10 W/m^2·K. The surrounding temperature is $T_{sur} = T_\infty$ = 300 K and the effective radiative heat transfer coefficient is h_{rad} = 5 W/m^2·K. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

(a) Show the thermal resistance network that can describe heat transfer across the PV panel. Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

(b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si}.

(c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

Knowns:

t_g = 3mm

k_g = 1.5 W/m·K

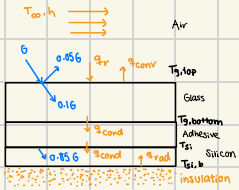
t_a = 0.1 mm

k_a = 1 W/m·K

Find: thermal resistance network (a), $T_{g,top}$ and T_{Si} (b), η and q''_{elec} (c)

Assumptions: steady state, 1D heat flow, constant material properties, insulated back

Schematic:



a)

Key assumptions:

Steady state, 1D heat flow, constant material properties, insulated back

$$R''_{conv} = \frac{1}{h} = \frac{1}{10} = 0.1 \ \frac{m^2 K}{W}$$

$$R''_{rad} = \frac{1}{h_{rad}} = \frac{1}{5} = 0.2 \ \frac{m^2 K}{W}$$

$$R''_{g,cond} = \frac{L_g}{k_g} = \frac{0.003}{1.5} = 0.002 \ \frac{m^2 K}{W}$$

$$R''_{a,cond} = \frac{L_a}{k_a} = \frac{0.0001}{1} = 1\times10^{-4} \ \frac{m^2 K}{W}$$

$$R''_c = 1\times10^{-4} \ \frac{m^2 K}{W}$$

b) Silicon: $q''_{cond} = q''_{Si}$

$$q''_{cond} = \frac{T_{Si} - T_{g,top}}{R''_{cond}}$$

$$q''_{Si} = 0.85G - 0.85G\eta$$

$$= 0.85G(1-\eta)$$

$$= 0.85G(1 - 0.553 + 0.001T_{Si})$$

$$= 0.85(800)(0.447 + 0.001T_{Si})$$

$$q''_{Si} = 303.96 + 0.68T_{Si}$$

$$\frac{T_{Si} - T_{g,top}}{0.0022} = 303.96 + 0.68T_{Si}$$

$$T_{Si} - T_{g,top} = 0.669 + 1.496\times10^{-3}T_{Si}$$

$$T_{g,top} = 0.999T_{Si} - 0.669$$

Glass top: $q''_{conv} + q''_{rad} = q''_{g,abs} + q''_{cond}$

$$(h + h_{rad})(T_{g,top} - 300) = 0.10G + q''_{cond}$$

$$15T_{g,top} - 4500 = 80 + \frac{T_{Si} - T_{g,top}}{0.0022}$$

$$0.033T_{g,top} - 10.076 = T_{Si} - T_{g,top}$$

$$1.033T_{g,top} - 10.076 = T_{Si}$$

$$T_{g,top} = 0.999(1.033T_{g,top} - 10.076) - 0.668$$

$$-0.0320T_{g,top} = -10.734$$

$$\boxed{T_{g,top} = 341.12 \text{ K}}$$

$$\boxed{T_{Si} = 1.033(341.12) - 10.076 = 342.30 \text{ K}}$$

c) $\eta = 0.553 - 0.001T_{Si}$

$$= 0.553 - 0.001(342.30)$$

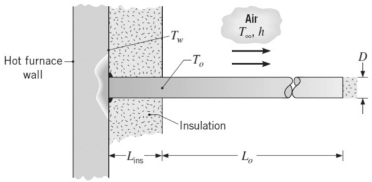
$$\boxed{\Rightarrow \eta = 0.21}$$

$$q''_{elec} = \eta q''_{Si,abs}$$

$$= 0.21(680)$$

$$\Rightarrow q''_{elec} = 14$$

2. (30 pt) A rod of diameter $D = 10$ mm and thermal conductivity $k = 80$ W/m·K protrudes normally from a furnace wall that is at $T_w = 200$ °C and is covered by insulation of thickness $L_{ins} = 200$ mm. The rod is welded to the furnace wall and is used as a hanger for supporting instrumentation cables. The rod has an exposed length L_o and its tip is well insulated. To avoid damaging the cables, the temperature of the rod at its exposed surface, T_o, must be maintained below a specified operating limit of $T_{max} = 100$ °C. The ambient air temperature is $T_\infty = 20$ °C and convective heat transfer coefficient is $h = 20$ W/m²·K.



(a) Derive an expression for the exposed surface temperature T_o as a function of the prescribed thermal geometrical parameters.

(b) If exposed length L_o is 200 mm, calculate the corresponding exposed surface temperature T_o. Does the exposed length meet the specified operating limit?

(c) Calculate the minimum exposed length L_o that ensures T_o below the specified operating limit.

Knowns: $D = 0.010\,m$, $K = 80\ \frac{W}{mK}$, $L_{ins} = 0.200\,m$

$T_w = 200\,^\circ C$

$T_\infty = 20\,^\circ C$, $h = 20\ \frac{W}{m^2 K}$

$T_{max} = 100\,^\circ C$

In (b): $L_o = 0.200\,m$

Find: Expression for T_o (a),

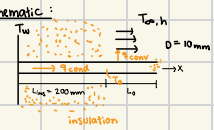
T_o for given L_o, check $T_o \leq T_{max}$ (b),

Minimum L_o with $T_o \leq T_{max}$ (c)

Assumptions: steady state, 1D conduction, uniform fin cross-section, insulated fin tip

fin assumption for exposed rod

Schematic:



Analysis:

a) Conduction: $q = \frac{KA_c}{L_{ins}}(T_w - T_o)$

Removed: $q_f = M\tanh(mL_o)$

$\sqrt{hPKA_c}\,(T_o - T_\infty)\tanh\left(\sqrt{\frac{hP}{KA_c}}\,L_o\right)$

$= \sqrt{hPKA_c}\,(T_o - T_\infty)\tanh(mL_o)$

Insulated tip: $\left.\frac{dT}{dx}\right|_{L_o} = 0$

$$\frac{1}{L_{ins}}(T_w - T_o) = m(T_o - T_\infty)\tanh(mL_o)$$

$$T_w - T_o = (T_o - T_\infty)L_{ins}\,m\tanh(mL_o)$$

$$T_o = \frac{T_w + L_{ins}\,m\tanh(mL_o)\,T_\infty}{1 + L_{ins}\,m\tanh(mL_o)}$$

$$m = \sqrt{\frac{h\pi D}{K(\pi D^2)/4}} = \sqrt{\frac{4h}{KD}}$$

$$\boxed{T_o = \frac{T_w + L_{ins}\sqrt{\frac{4h}{KD}}\tanh\left(\sqrt{\frac{4h}{KD}}\,L_o\right)T_\infty}{1 + L_{ins}\sqrt{\frac{4h}{KD}}\tanh\left(\sqrt{\frac{4h}{KD}}\,L_o\right)}}$$

b) $L_{ins}\sqrt{\frac{4h}{KD}}\tanh\left(\sqrt{\frac{4h}{KD}}\,L_o\right) = 0.2\sqrt{\frac{4(20)}{80(0.01)}}\tanh\left(\sqrt{\frac{4(20)}{80(0.01)}}\,0.2\right) = 1.928$

$$T_o = \frac{200 + 1.928(20)}{1 + 1.928}$$

$$\boxed{\Rightarrow T_o = 81.48\,^\circ C \leq 100\,^\circ C \quad \text{Meets limit}}$$

c) Set $T_o = T_{max}$

$$T_{max} = \frac{T_w + L_{ins}\sqrt{\frac{4h}{KD}}\tanh\left(\sqrt{\frac{4h}{KD}}\,L_o\right)T_\infty}{1 + L_{ins}\sqrt{\frac{4h}{KD}}\tanh\left(\sqrt{\frac{4h}{KD}}\,L_o\right)}$$

$$L_{ins}\sqrt{\frac{4h}{KD}}\tanh\left(\sqrt{\frac{4h}{KD}}\,L_o\right) = \frac{T_w - T_{max}}{T_{max} - T_\infty}$$

$$= \frac{200 - 100}{100 - 20}$$

$$2\tanh(10L_o) = 1.25$$

$$L_o = \frac{1}{10}\tanh^{-1}\left(\frac{1.25}{2}\right)$$

$$\boxed{\Rightarrow L_o = 0.0733\,m = 73.3\,mm}$$

3. (30 pt) A square pin fin of uniform cross-section area is fabricated of copper (k = 400 W/m·K). The fin width is s = 10 mm and the fin is exposed to air flow with convective heat transfer coefficient h = 100 W/m^2·K. It is known that the fin efficiency is η_f = 0.6. The fin has a convection tip.

(a) Calculate the fin length L.

(b) Calculate the thermal resistance of the fin R_f and the fin effectiveness ε_f.

Knowns: $K = 400 \frac{W}{mK}$

$s = 0.01m \Rightarrow A = 0.01^2 = 1\times10^{-4} m^2$, $P = 4(0.01) = 0.04$

$h = 100 \frac{W}{m^2K}$

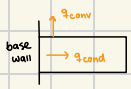
$\eta_f = 0.6$

Find: Fin length L (a), fin thermal resistance R_f and effectiveness ε_f

Assumptions: steady state, 1D conduction, uniform cross-section, constant K and h, corrected length approximation

fin assumption

Schematic:



Analysis:

a) $L_c = L + \frac{s}{4}$

$\eta_f = \frac{\tanh(mL_c)}{mL_c}$, $m = \sqrt{\frac{hP}{KA_c}}$

$\frac{\tanh(mL_c)}{mL_c} = 0.6$

$mL_c \approx 1.5$

$L_c = \frac{1.5}{\sqrt{\frac{100(0.04)}{400(10^{-4})}}} = 0.15m$

$L = L_c - \frac{A_c}{P} = 0.15 - \frac{0.01}{4} = 0.1475$

$\Rightarrow L = 0.148m = 148mm$

b) $q_f = \eta_f h A_f (T_b - T_\infty)$

$\frac{1}{\eta_f h A_f} = \frac{T_b - T_\infty}{q_f}$

$R_f = \frac{1}{\eta_f h A_f}$

$= \frac{1}{\eta_f h (PL + A_c)}$

$= \frac{1}{0.6(100)(0.04(0.148 + 10^{-4}))}$

$R_f = 2.78 \frac{K}{W}$

$\varepsilon_f = \frac{q_f}{q_{nf}}$

$= \sqrt{\frac{KP}{hA_c}} \tanh(mL_c)$

$= \sqrt{\frac{400(0.04)}{100(10^{-4})}} \tanh(1.5)$

$\varepsilon_f = 36.2$